

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electronics and Electrical Communication Engineering

2. Subject Name: Systems and Control L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: -Network Theory (

4. Syllabus and reference books:

Syllabus:

Modeling: Historical overview, transfer characteristics, tracking and disturbance rejection, transfer function, Signal flow graph, time and frequency domain behavior of LTI systems.; Feedback: Open loop and closed loop systems, stability analysis, poles and zeros, RH criteria, Root locus, Bode plot, Nyquist plot, Nyquist criteria, gain margin, phase margin.; State space: Nonlinear dynamics, operating point and linearization, Eigen analysis, Matrix exponential, stability, controllability, observability, reachability, realizability, sensitivity, observer, Lyapunov stability, canonical form and minimal realization.; Controller synthesis: PI, PID, Lead-lag, pole placement controllers.; Robust control: Frequency domain techniques, time domain techniques.; Optimal control: Pontryagin principle, Calculus of variation, HJB equation, dynamic programming, Linear quadratic regulator, Linear quadratic Gaussian.; Model inference: Parameter estimation, observer design, model predictive control, adaptive control.

Reference Books:

1. Lecture Notes on control system theory and design, Tamer Basar, Sean P. Meyn, and William R. Perkins, 2010.
2. Linear systems theory, J. Hespanha, Princeton Press, 2018.
3. Linear Control Theory: Structure, Robustness, and Optimization, Bhattacharyya, Datta, and Keel, CRC Press, 2009.
4. Linear Optimal Control, Anderson and Moore, Prentice Hall, 1990
5. Adaptive Control: Stability, Convergence and Robustness, S. Sastry and M. Bodson, Prentice Hall, 1989

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Modeling: Historical overview, transfer characteristics, tracking and disturbance rejection, transfer function, Signal flow graph, time and frequency domain behavior of LTI systems.	6
2.	Feedback: Open loop and closed loop systems, stability analysis, poles and zeros, RH criteria, Root locus, Bode plot, Nyquist plot, Nyquist criteria, gain margin, phase margin.	6
3.	State space: Nonlinear dynamics, operating point and linearization, Eigen analysis, Matrix exponential, stability, controllability, observability, reachability, realizability, sensitivity, Lyapunov stability, Canonical form and minimal realization.	8
4.	Controller synthesis: PI, PID, Lead-lag, pole placement controllers.	8
5.	Robust control: Frequency domain techniques, time domain techniques.	6
6.	Optimal control: Pontryagin principle, Calculus of variation, HJB equation, dynamic programming, Linear quadratic regulator, Linear quadratic Gaussian.	8
7.	Model inference: Parameter estimation, observer design, model predictive control, adaptive control.	6
Total number of hours		48